

# VEDANT KAPIL

AI Engineer | Machine Learning | Generative AI | Python Developer | Data Analytics  
India | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

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## PROFESSIONAL SUMMARY

AI Engineer with hands-on project experience building applications using Large Language Models, Retrieval-Augmented Generation (RAG), Computer Vision, and Machine Learning. Skilled in Python, LangChain, FastAPI, and data analytics, with a focus on developing practical, production-quality AI systems and continuously expanding expertise in AI engineering and backend development.

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## TECHNICAL SKILLS

**Programming:** Python, SQL

**Data Analytics:** NumPy, Pandas, Matplotlib, Scikit-learn, Data Cleaning, Exploratory Data Analysis, Feature Engineering, Data Visualization

**Machine Learning:** Supervised Learning, Regression, Classification, Model Evaluation, Data Preprocessing

**Generative AI:** Large Language Models (LLMs), LangChain, RAG, Prompt Engineering, Embeddings, Sentence Transformers, Ollama, ChromaDB, AI Agents

**Computer Vision:** YOLO, OpenCV, Roboflow

**Backend:** FastAPI, REST APIs, JSON, HTTP APIs

**Developer Tools:** Git, GitHub, VS Code, Linux

**Currently Learning:** Docker, AWS, PyTorch, Deep Learning

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## PROJECTS

**AI Portfolio Website** | *React, TypeScript, GitHub API, GitHub Pages, REST APIs*

- Designed and developed a modern portfolio website using React and TypeScript.
- Integrated the GitHub API to automatically display repositories, keeping project listings up to date without manual edits.
- Implemented an admin dashboard enabling portfolio content updates through GitHub using Personal Access Tokens.
- Configured GitHub Pages deployment to automatically publish updates whenever repository content changes.

**Amazon Product RAG** | *Python, LangChain, ChromaDB, Ollama, Sentence Transformers, RAG*

- Developed a Retrieval-Augmented Generation (RAG) application capable of answering product-related questions.
- Converted product data into embeddings using Sentence Transformers for semantic search.
- Stored and indexed embeddings in ChromaDB for efficient retrieval.
- Retrieved relevant context using LangChain and generated responses locally using Ollama.

**AI Chatbot** | *Python, LangChain, Ollama, Streamlit*

- Built a conversational AI chatbot using LangChain and Ollama with local response generation.
- Implemented streaming responses and maintained chat history for continuous conversation context.
- Created an interactive user interface using Streamlit.

**Traffic Detection System** | *YOLO, OpenCV, Roboflow, Python*

- Developed a real-time object detection system using YOLO for identifying traffic elements.
- Trained and evaluated the detection model using Roboflow datasets.
- Applied OpenCV for image processing and video frame handling.

**Data Analytics Projects** | *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

- Performed exploratory data analysis (EDA) to uncover trends and data quality issues.
- Built machine learning models using Scikit-learn for regression and classification tasks.
- Applied data cleaning and preprocessing techniques using Pandas and NumPy.
- Created visualizations using Matplotlib to communicate insights clearly.

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## EDUCATION

**Bachelor of Engineering, Computer Engineering**